परीक्षा प्लस

उत्तर प्रदेश लोक सेवा आयोग

LT GRADE

OPERATING SYSTEM

परीक्षा

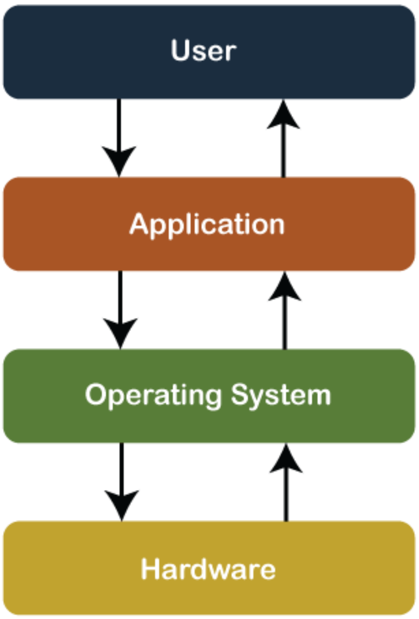
- नवीनतम पाठ्यक्रम पर आधारित
- विगत वर्षों के प्रश्न पत्र के विश्लेषण पर आधारित
- सारगर्भित एवं गुणवत्तापूर्ण
- पूर्णतः संशोधित

Operating System

An **Operating System** can be defined as an **interface between user and hardware**. It is responsible for the execution of all the processes, Resource Allocation, CPU management, File Management and many other tasks.

The purpose of an operating system is to provide an environment in which a user can execute programs in convenient and efficient manner.

.



Functions of Operating System

- **Resource Management:** When parallel accessing happens in the OS means when multiple users are accessing the system the OS works as Resource Manager, Its responsibility is to provide hardware to the user. It decreases the load in the system.
- **Process Management:** It includes various tasks like **scheduling and termination** of the process. It is done with the help of CPU Scheduling algorithms.
- **Storage Management:** The **file system** mechanism used for the management of the storage. **NIFS**, **CIFS**, **CFS**, **NFS**, etc. are some file systems. All the data is stored in various tracks of Hard disks that are all managed by the storage manager. It included **Hard Disk**.

Memory Management: Refers to the management of primary memory. The operating system has to keep track of how much memory has been used and by whom. It has to decide which process needs memory space and how much. OS also has to allocate and deallocate the memory space.

Security/Privacy Management: Privacy is also provided by the Operating system using passwords so that unauthorized applications can't access programs or data. For example, Windows uses **Kerberos** authentication to prevent unauthorized access to data.

Components of an Operating Systems

There are two basic components of an Operating System.

- Shell
- Kernel

Shell

- Shell is the outermost layer of the Operating System and it handles the interaction with the user.
- The main task of the Shell is the management of interaction between the User and OS.
- Shell provides better communication with the user and the Operating System Shell does it by giving proper input to the user it also interprets input for the OS and handles the output from the OS.
- It works as a way of communication between the User and the OS.

Kernel

- The kernel is one of the components of the Operating System which works as a core component.

- The rest of the components depends on Kernel for the supply of the important services that are provided by the Operating System.
- The kernel is the primary interface between the Operating system and Hardware.

Functions of Kernel

The following functions are to be performed by the Kernel.

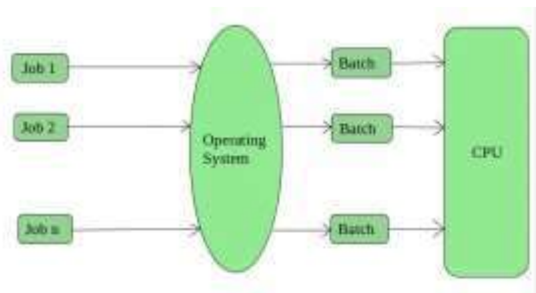
- It helps in controlling the System Calls.
- It helps in I/O Management.
- It helps in the management of applications, memory, etc.

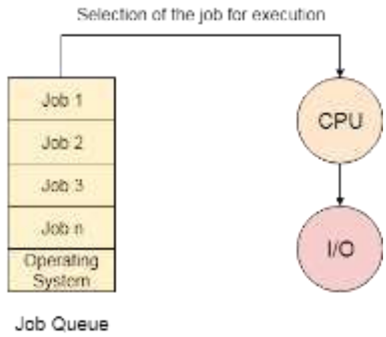
Types of Operating System

1. Batch Operating System
2. Multiprogramming Operating System
3. Time-Sharing Operating Syste
4. Real-Time Operating System
5. Distributed Operating System
6. Clusterd Operating system
7. Multiprocessing Operating System
8. Embedded Operating System

Batch Operating System

- This type of operating system does not interact with the computer directly.
- There is an operator which takes similar jobs having the same requirement and groups them into batches.
- It is the responsibility of the operator to sort jobs with similar needs.





There are five jobs J1, J2, J3, J4, and J5, present in the batch. If the execution time of J1 is very high, then the other four jobs will never be executed, or they will have to wait for a very long time. Hence the other processes get starved.

Advantages of Batch Operating System

- It is very difficult to guess or know the time required for any job to complete. Processors of the batch systems know how long the job would be when it is in the queue.
- Multiple users can share the batch systems.
- The idle time for the batch system is very less.
- It is easy to manage large work repeatedly in batch systems.

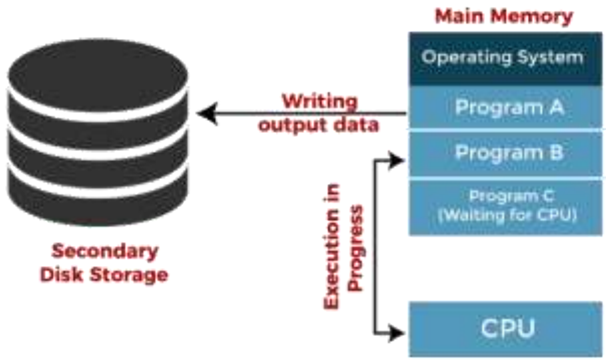
Disadvantages of Batch Operating System

- The computer operators should be well known with batch systems.
- Batch systems are hard to debug.
- It is sometimes costly.
- The other jobs will have to wait for an unknown time if any job fails.

Examples of Batch Operating Systems: Payroll Systems, Bank Statements, etc.

Multi-Programming Operating System

Multiprogramming Operating Systems can be simply illustrated as more than one program is present in the main memory and any one of them can be kept in execution. This is basically used for better execution of resources.



Jobs in multiprogramming system

Advantages of Multi-Programming Operating System

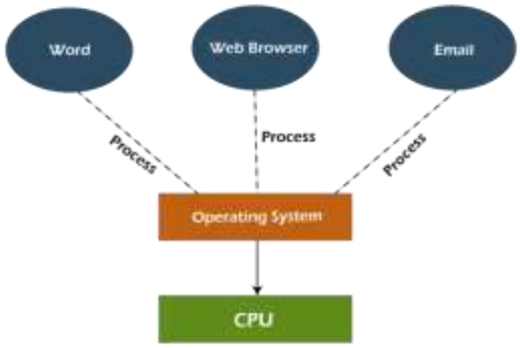
- Multi Programming increases the Throughput of the System.
- It helps in reducing the response time.

Disadvantages of Multi-Programming Operating System

- There is not any facility for user interaction of system resources with the system.

➢ Time-Sharing Operating Systems

- Each task is given some time to execute so that all the tasks work smoothly.
- Each user gets the time of the CPU as they use a single system.
- These systems are also known as Multitasking Systems.
- The task can be from a single user or different users also.
- The time that each task gets to execute is called quantum.
- After this time interval is over OS switches over to the next task.



Advantages of Time-Sharing OS

- Each task gets an equal opportunity.
- Fewer chances of duplication of software.
- CPU idle time can be reduced.
- Resource Sharing: Time-sharing systems allow multiple users to share hardware resources such as the CPU, memory, and peripherals, reducing the cost of hardware and increasing efficiency.
- Improved Productivity: Time-sharing allows users to work concurrently, thereby reducing the waiting time for their turn to use the computer. This increased productivity translates to more work getting done in less time.
- Improved User Experience: Time-sharing provides an interactive environment that allows users to communicate with the computer in real time, providing a better user experience than batch processing.

Disadvantages of Time-Sharing OS

- Reliability problem.
- One must have to take care of the security and integrity of user programs and data.
- Data communication problem.
- High Overhead: Time-sharing systems have a higher overhead than other operating systems due to the need for scheduling, context switching, and other overheads that come with supporting multiple users.
- Complexity: Time-sharing systems are complex and require advanced software to manage multiple users simultaneously. This complexity increases the chance of bugs and errors.

- Security Risks: With multiple users sharing resources, the risk of security breaches increases. Time-sharing systems require careful management of user access, authentication, and authorization to ensure the security of data and software.

➢ Real-Time Operating System

These types of OSs serve real-time systems. The time interval required to process and respond to inputs is very small. This time interval is called **response time**. **Real-time systems** are used when there are time requirements that are very strict like missile systems, air traffic control systems, robots, etc.

Types of Real-Time Operating Systems

- **Hard Real-Time Systems:** Hard Real-Time OSs are meant for applications where time constraints are very strict and even the shortest possible delay is not acceptable. These systems are built for saving life like automatic parachutes or airbags which are required to be readily available in case of an accident. Virtual memory is rarely found in these systems.
- **Soft Real-Time Systems:** These OSs are for applications where time-constraint is less strict.

Advantages of RTOS

- **Maximum Consumption:** Maximum utilization of devices and systems, thus more output from all the resources.
- **Task Shifting:** The time assigned for shifting tasks in these systems is very less. For example, in older systems, it takes about 10 microseconds in shifting from one task to another, and in the latest systems, it takes 3 microseconds.
- **Focus on Application:** Focus on running applications and less importance on applications that are in the queue.
- Real-time **operating system in** the **embedded system:** Since the size of programs is small, RTOS can also be used in embedded systems like in transport and others.
- **Error Free:** These types of systems are error-free.
- **Memory Allocation:** Memory allocation is best managed in these types of systems.

Disadvantages of RTOS

- **Limited Tasks:** Very few tasks run at the same time and their concentration is very less on a few applications to avoid errors.
- **Use heavy system resources:** Sometimes the system resources are not so good and they are expensive as well.
- **Complex Algorithms:** The algorithms are very complex and difficult for the designer to write on.
- **Device driver and interrupt signals:** It needs specific device drivers and interrupts signal to respond earliest to interrupts.
- **Thread Priority:** It is not good to set thread priority as these systems are very less prone to switching tasks.

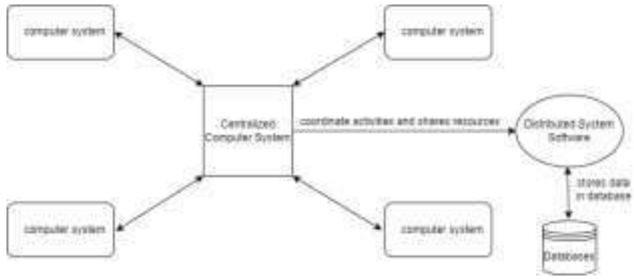
Examples of Real-Time Operating Systems are Scientific experiments, medical imaging systems, industrial control systems, weapon systems, robots, air traffic control systems, etc.

➢ Distributed Operating System

- Distributed System is a collection of autonomous computer systems that are physically separated but are connected by a centralized computer network that is equipped with distributed system software.
- The autonomous computers will communicate among each system by sharing resources and files and performing the tasks assigned to them.

Example of Distributed System:

Any Social Media can have its Centralized Computer Network as its Headquarters and computer systems that can be accessed by any user and using their services will be the Autonomous Systems in the Distributed System Architecture.



- **Distributed System Software:** This Software enables computers to coordinate their activities and to share the resources such as Hardware, Software, Data, etc.
- **Database:** It is used to store the processed data that are processed by each Node/System of the Distributed systems that are connected to the Centralized network.

Advantages of Distributed System:

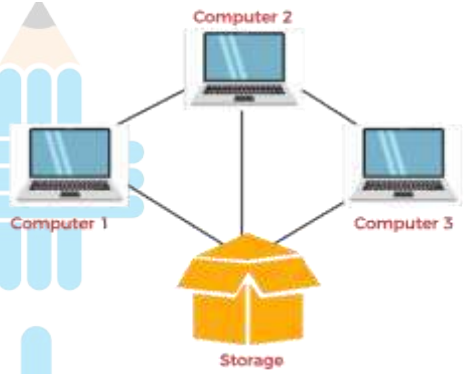
- Applications in Distributed Systems are Inherently Distributed Applications.
- Information in Distributed Systems is shared among geographically distributed users.
- Resource Sharing (Autonomous systems can share resources from remote locations).
- It has a better price performance ratio and flexibility.
- It has shorter response time and higher throughput.
- It has higher reliability and availability against component failure.
- It has extensibility so that systems can be extended in more remote locations and also incremental growth.

Applications Area of Distributed System:

- **Finance and Commerce:** Amazon, eBay, Online Banking, E-Commerce websites.
- **Information Society:** Search Engines, Wikipedia, Social Networking, Cloud Computing.
- **Cloud Technologies:** AWS, Salesforce, Microsoft Azure, SAP.
- **Entertainment:** Online Gaming, Music, youtube.
- **Healthcare:** Online patient records, Health Informatics.
- **Education:** E-learning.
- **Transport and logistics:** GPS, Google Maps.
- **Environment Management:** Sensor technologies.

Clustered Operating System

- Cluster systems are similar to parallel systems because both systems use multiple CPUs.
- The primary difference is that clustered systems are made up of two or more independent systems linked together.
- They have independent computer systems and a shared storage media, and all systems work together to complete all tasks.

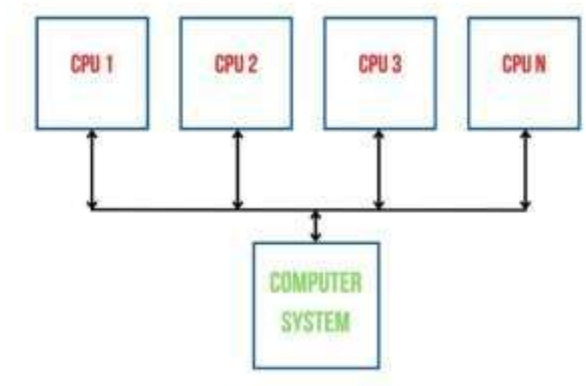


Clustered System

➢ Multiprocessing Operating System

- Multiprocessor operating systems are used in operating systems to boost the performance of multiple CPUs within a single computer system.
- Multiple CPUs are linked together so that a job can be divided and executed more quickly.
- When a job is completed, the results from all CPUs are compiled to provide the final output. Jobs were required to share main memory, and they may often share other system resources.
- Multiple CPUs can be used to run multiple tasks at the same time, for example, UNIX.

MULTIPROCESSING



Advantages of Multi-Processing Operating System

- It increases the throughput of the system.

Disadvantages of Multi-Processing Operating System

- Due to the multiple CPU, it can be more complex and somehow difficult to understand.

Embedded Operating System

- The Embedded operating system is the specific purpose operating system used in the computer system's embedded hardware configuration.
- These operating systems are designed to work on dedicated devices like automated teller machines (ATMs), airplane systems, digital home assistants, and the internet of things (IoT) devices.

Function of Windows

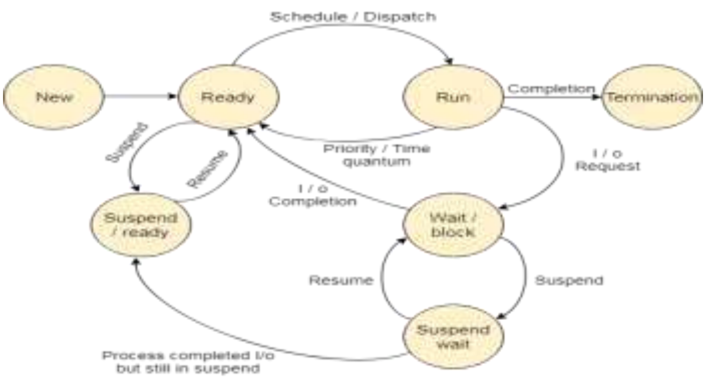
Process Management

If the operating system supports multiple users then services under this are very important. In this regard, operating systems have to keep track of all the completed processes, Schedule them, and dispatch them one after another. But the user should feel that he has full control of the CPU.

States of Process

A process is in one of the following states:

1. **New:** Newly Created Process (or) being-created process.
2. **Ready:** After the creation process moves to the Ready state, i.e. the process is ready for execution.
3. **Run:** Currently running process in CPU (only one process at a time can be under execution in a single processor)
4. **Wait (or Block):** When a process requests I/O access.
5. **Complete (or Terminated):** The process completed its execution.
6. **Suspended Ready:** When the ready queue becomes full, some processes are moved to a suspended ready state
7. **Suspended Block:** When the waiting queue becomes full.



1. New

A program which is going to be picked up by the OS into the main memory is called a new process.

2. Ready

Whenever a process is created, it directly enters in the ready state, in which, it waits for the CPU to be assigned. The OS picks the new processes from the secondary memory and put all of them in the main memory.

The processes which are ready for the execution and reside in the main memory are called ready state processes. There can be many processes present in the ready state.

3. Running

- One of the processes from the ready state will be chosen by the OS depending upon the scheduling algorithm.
- Hence, if we have only one CPU in our system, the number of running processes for a particular time will always be one.
- If we have n processors in the system then we can have n processes running simultaneously.

4. Block or wait

- From the Running state, a process can make the transition to the block or wait state depending upon the scheduling algorithm or the intrinsic behavior of the process.
- When a process waits for a certain resource to be assigned or for the input from the user then the OS move this process to the block or wait state and assigns the CPU to the other processes.

5. Completion or termination

When a process finishes its execution, it comes in the termination state. All the context of the process (Process Control Block) will also be deleted the process will be terminated by the Operating system.

6. Suspend ready

- A process in the ready state, which is moved to secondary memory from the main memory due to lack of the resources (mainly primary memory) is called in the suspend ready state.
- If the main memory is full and a higher priority process comes for the execution then the OS have to make the room for the process in the main memory by throwing the lower priority process out into the secondary memory.
- The suspend ready processes remain in the secondary memory until the main memory gets available.

7. Suspend wait

- Instead of removing the process from the ready queue, it's better to remove the blocked process which is waiting for some resources in the main memory.
- Since it is already waiting for some resource to get available hence it is better if it waits in the secondary memory and make room for the higher priority process.
- These processes complete their execution once the main memory gets available and their wait is finished.

Process Schedulers

Operating system uses various schedulers for the process scheduling described below.

1. Long term scheduler

- Long term scheduler is also known as job scheduler.
- It chooses the processes from the pool (secondary memory) and keeps them in the ready queue maintained in the primary memory.
- Long Term scheduler mainly controls the degree of Multiprogramming.
- The purpose of long term scheduler is to choose a perfect mix of IO bound and CPU bound processes among the jobs present in the pool.
- If the job scheduler chooses more IO bound processes then all of the jobs may reside in the

blocked state all the time and the CPU will remain idle most of the time.

- This will reduce the degree of Multiprogramming. Therefore, the Job of long term scheduler is very critical and may affect the system for a very long time.

2. Short term scheduler

- Short term scheduler is also known as CPU scheduler. It selects one of the Jobs from the ready queue and dispatch to the CPU for the execution.
- A scheduling algorithm is used to select which job is going to be dispatched for the execution.
- The Job of the short term scheduler can be very critical in the sense that if it selects job whose CPU burst time is very high then all the jobs after that, will have to wait in the ready queue for a very long time.
- This problem is called starvation which may arise if the short term scheduler makes some mistakes while selecting the job.

3. Medium term scheduler

- Medium term scheduler takes care of the swapped out processes.If the running state processes needs some IO time for the completion then there is a need to change its state from running to waiting.
- Medium term scheduler is used for this purpose.
- It removes the process from the running state to make room for the other processes.
- Such processes are the swapped out processes and this procedure is called swapping. The medium term scheduler is responsible for suspending and resuming the processes.
- It reduces the degree of multiprogramming.
- The swapping is necessary to have a perfect mix of processes in the ready queue.

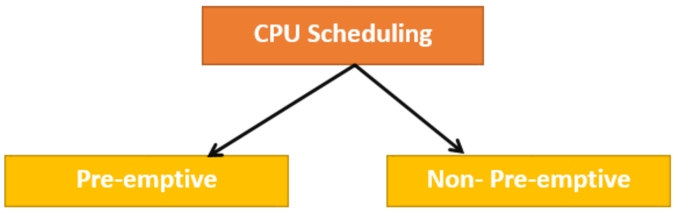
Context Switching: The process of saving the context of one process and loading the context of another process is known as Context Switching. In simple terms, it is like loading and unloading the process from the running state to the ready state.

CPU Scheduling

- **In the uniprogrammming systems** like MS DOS, when a process waits for any I/O operation to be done, the CPU remains idol.
- This is an overhead since it wastes the time and causes the problem of starvation.
- However, In Multiprogramming systems, the CPU doesn't remain idle during the waiting time of the Process and it starts executing other processes.
- Operating System has to define which process the CPU will be given.
- **In Multiprogramming systems**, the Operating system schedules the processes on the CPU to have the maximum utilization of it and this procedure is called **CPU scheduling**.
- The Operating System uses various scheduling algorithm to schedule the processes.

Types of CPU Scheduling

Here are **two kinds** of Scheduling methods:



➢ **Preemptive Scheduling**

- In Preemptive Scheduling, the tasks are mostly assigned with their priorities.
- Sometimes it is important to run a task with a higher priority before another lower priority task, even if the lower priority task is still running.

- The lower priority task holds for some time and resumes when the higher priority task finishes its execution.

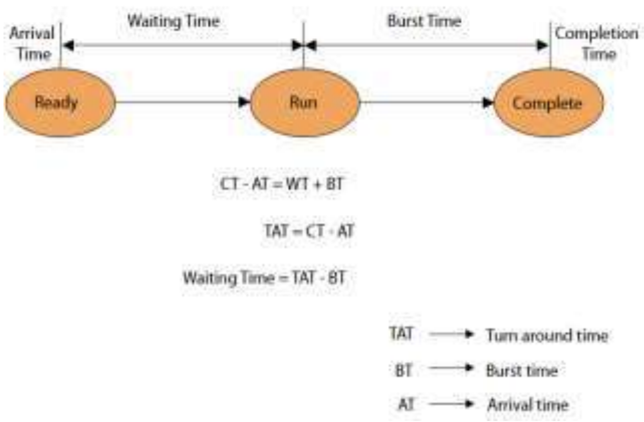
➢ **Non-Preemptive Scheduling**

- In this type of scheduling method, the CPU has been allocated to a specific process.
- The process that keeps the CPU busy will release the CPU either by switching context or terminating.
- It is the only method that can be used for various hardware platforms. That's because it doesn't need special hardware (for example, a timer) like preemptive scheduling.

The Purpose of a Scheduling algorithm

1. Maximum CPU utilization
2. Fare allocation of CPU
3. Maximum throughput
4. Minimum turnaround time
5. Minimum waiting time
6. Minimum response time

Various Times related to the Process



1. Arrival Time

The time at which the process enters into the ready queue is called the arrival time.

2. Burst Time

The total amount of time required by the CPU to execute the whole process is called the Burst Time. This does not include the waiting time. It is confusing to calculate the execution time for a process even before executing it hence the scheduling problems based on the burst time cannot be implemented in reality.

3. Completion Time

The Time at which the process enters into the completion state or the time at which the process completes its execution, is called completion time.

4. Turnaround time

The total amount of time spent by the process from its arrival to its completion, is called Turnaround time.

5. Waiting Time

The Total amount of time for which the process waits for the CPU to be assigned is called waiting time.

6. Response Time

The difference between the arrival time and the time at which the process first gets the CPU is called Response Time

Types of CPU scheduling Algorithm

There are mainly six types of process scheduling algorithms

1. First Come First Serve (FCFS)
2. Shortest-Job-First (SJF) Scheduling
3. Shortest Remaining Time
4. Priority Scheduling
5. Round Robin Scheduling

1. First Come First Serve

It is the simplest algorithm to implement. The process with the minimal arrival time will get the CPU first. The lesser the arrival time, the sooner will the process gets the CPU. It is the non-preemptive type of scheduling.

2. Round Robin

In the Round Robin scheduling algorithm, the OS defines a time quantum (slice). All the processes will get executed in the cyclic way. Each of the process will get the CPU for a small amount of time (called time quantum) and then get back to the ready queue to wait for its next turn. It is a preemptive type of scheduling

3. Shortest Job First

The job with the shortest burst time will get the CPU first. The lesser the burst time, the sooner will the process get the CPU. It is the non-preemptive type of scheduling.

4. Shortest remaining time first

It is the preemptive form of SJF. In this algorithm, the OS schedules the Job according to the remaining time of the execution.

5. Priority based scheduling

In this algorithm, the priority will be assigned to each of the processes. The higher the priority, the sooner will the process get the CPU. If the priority of the two processes is same then they will be scheduled according to their arrival time

System Calls in Operating System

- A system call is a method for a computer program to request a service from the kernel of the operating system on which it is running.
- A system call is a method of interacting with the operating system via programs. A system call is a request from computer software to an operating system's kernel.
- The **Application Program Interface (API)** connects the operating system's functions to user programs. It acts as a link between the operating system and a process, allowing user-level programs to request operating system services. The kernel system can only be accessed using system calls. System calls are required for any programs that use resources.

Memory Management

In a multiprogramming computer, the Operating System resides in a part of memory, and the rest is used by multiple processes. The task of subdividing the memory among different processes is called Memory Management. Memory management is a method in the operating system to manage operations between main memory and disk during process execution. The main aim of memory management is to achieve efficient utilization of memory.

Why Memory Management is Required?

- Allocate and de-allocate memory before and after process execution.
- To keep track of used memory space by processes.
- To minimize fragmentation issues.
- To proper utilization of main memory.
- To maintain data integrity while executing of process.

Logical and Physical Address Space

- **Logical Address Space:** An address generated by the CPU is known as a "Logical Address". It is also known as a Virtual address. Logical address space can be defined as the size of the process. A logical address can be changed.
- **Physical Address Space:** An address seen by the memory unit (i.e the one loaded into the memory address register of the memory) is commonly known as a "Physical Address". A Physical address is also known as a Real address. The set of all physical addresses corresponding to these logical

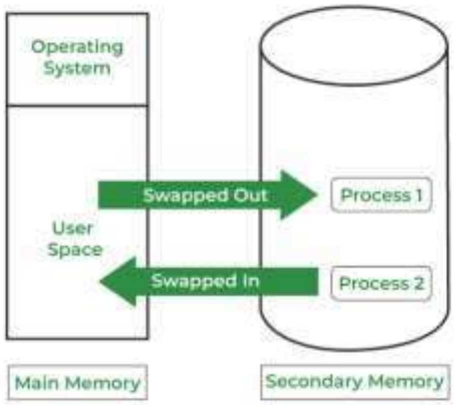
addresses is known as Physical address space. A physical address is computed by MMU. The run-time mapping from virtual to physical addresses is done by a hardware device Memory Management Unit(MMU). The physical address always remains constant

Swapping

When a process is executed it must have resided in memory. Swapping is a process of swapping a process temporarily into a secondary memory from the main memory, which is fast compared to secondary memory.

Swapping has been subdivided into two concepts: swap-in and swap-out.

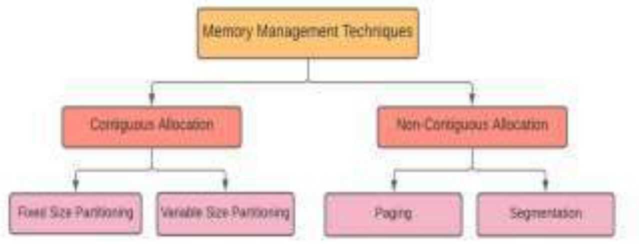
- Swap-out is a technique for moving a process from RAM to the hard disc.
- Swap-in is a method of transferring a program from a hard disc to main memory, or RAM.



Memory management Techniques:

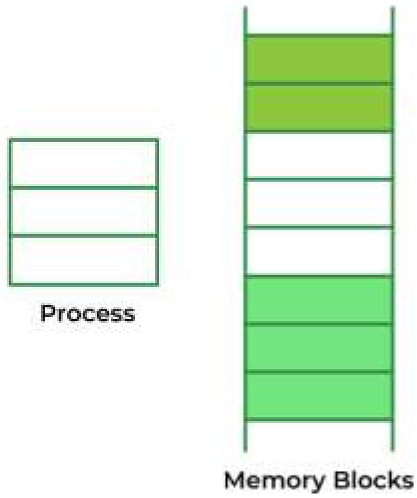
The Memory management Techniques can be classified into following main categories:

- Contiguous memory management schemes
- Non-Contiguous memory management schemes



➤ **Contiguous Memory Allocation**

Contiguous memory allocation is basically a method in which a single contiguous section/part of memory is allocated to a process or file needing it. Because of this all the available memory space resides at the same place together, which means that the freely/unused available memory partitions are not distributed in a random fashion here and there across the whole memory space



Contiguous Memory Allocation

- **Fixed/Static Partitioning** – In fixed partitioning, the memory is divided into fixed-size partitions, and each partition is assigned to a process. This technique is easy to implement but can result in wasted memory if a process does not fit perfectly into a partition.
- **Dynamic/Variable Partitioning** – In dynamic partitioning, the memory is divided into variablesize partitions, and each partition is assigned to a process. This technique is more

efficient as it allows the allocation of only the required memory to the process, but it requires more overhead to keep track of the available memory.

Advantages of Contiguous Memory Allocation

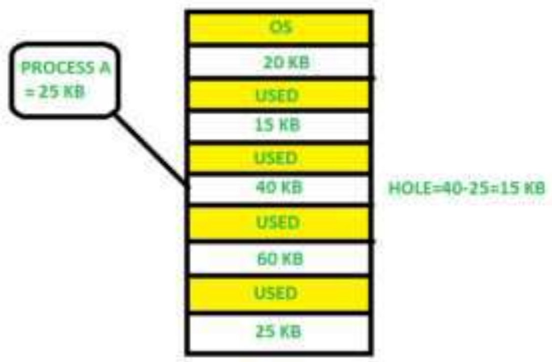
- Simplicity – Contiguous memory allocation is a relatively simple and straightforward technique for memory management. It requires less overhead and is easy to implement.
- Efficiency – Contiguous memory allocation is an efficient technique for memory management. Once a process is allocated contiguous memory, it can access the entire memory block without any interruption.
- Low fragmentation – Since the memory is allocated in contiguous blocks, there is a lower risk of memory fragmentation. This can result in better memory utilization, as there is less memory wastage.

Disadvantages of Contiguous Memory Allocation

- Limited flexibility – Contiguous memory allocation is not very flexible as it requires memory to be allocated in a contiguous block. This can limit the amount of memory that can be allocated to a process.
- Memory wastage – If a process requires a memory size that is smaller than the contiguous block allocated to it, there may be unused memory, resulting in memory wastage.
- Difficulty in managing larger memory sizes – As the size of memory increases, managing contiguous memory allocation becomes more difficult. This is because finding a contiguous block of memory that is large enough to allocate to a process becomes challenging.
- External Fragmentation – Over time, external fragmentation may occur as a result of memory allocation and deallocation, which may result in non – contiguous blocks of free memory scattered throughout the system.

First Fit

In the First Fit, the first available free hole fulfil the requirement of the process allocated.

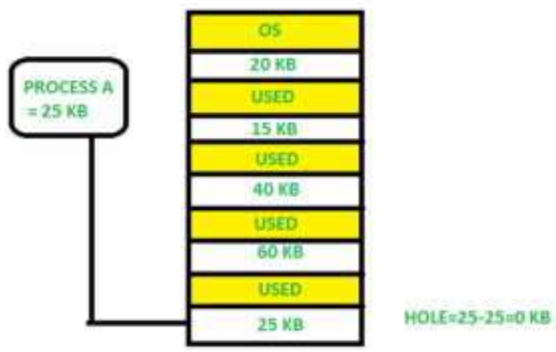


First Fit

Here, in this diagram, a 40 KB memory block is the first available free hole that can store process A (size of 25 KB), because the first two blocks did not have sufficient memory space.

Best Fit

In the Best Fit, allocate the smallest hole that is big enough to process requirements. For this, we search the entire list, unless the list is ordered by size.

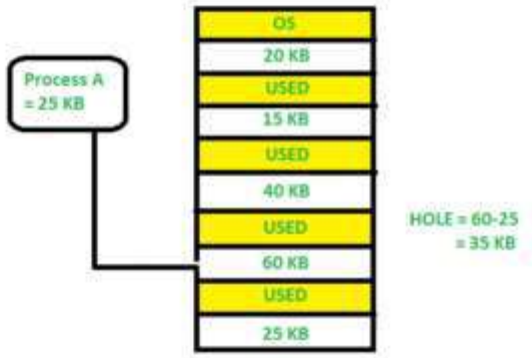


Best Fit

Here in this example, first, we traverse the complete list and find the last hole 25KB is the best suitable hole for Process A(size 25KB). In this method, memory utilization is maximum as compared to other memory allocation techniques.

Worst Fit

In the Worst Fit, allocate the largest available hole to process. This method produces the largest leftover hole.



Worst Fit

Here in this example, Process A (Size 25 KB) is allocated to the largest available memory block which is 60KB. Inefficient memory utilization is a major issue in the worst fit.

Fragmentation

Fragmentation is defined as when the process is loaded and removed after execution from memory, it creates a small free hole. These holes can not be assigned to new processes because holes are not combined or do not fulfill the memory requirement of the process. To achieve a degree of multiprogramming, we must reduce the waste of memory or fragmentation problems. In the operating systems two types of fragmentation:

1. Internal fragmentation:

Internal fragmentation occurs when memory blocks are allocated to the process more than their requested size. Due to this some unused space is left over and creating an internal fragmentation problem.

Example: Suppose there is a fixed partitioning used for memory allocation and the different sizes of blocks 3MB, 6MB, and 7MB space in memory. Now a new process p4 of size 2MB comes and demands a block of memory. It gets a memory block of 3MB but 1MB block of memory is a waste, and it can not be allocated to other processes too. This is called internal fragmentation.

2. External fragmentation:

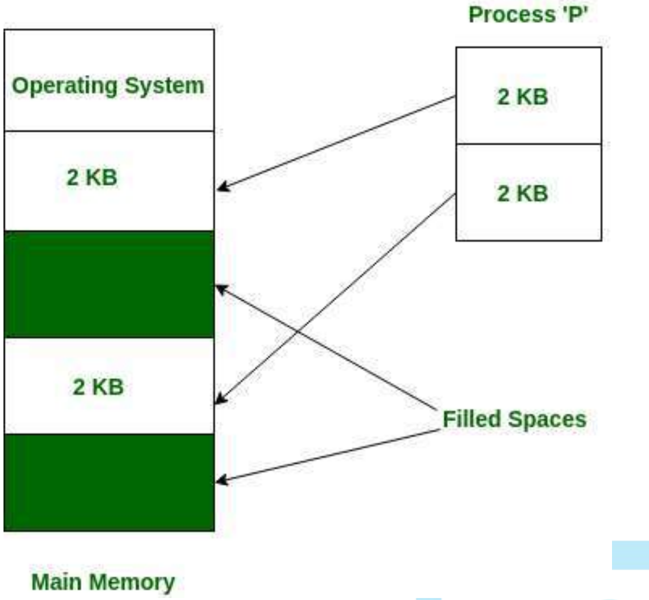
In External Fragmentation, we have a free memory block, but we can not assign it to a process because blocks are not contiguous.

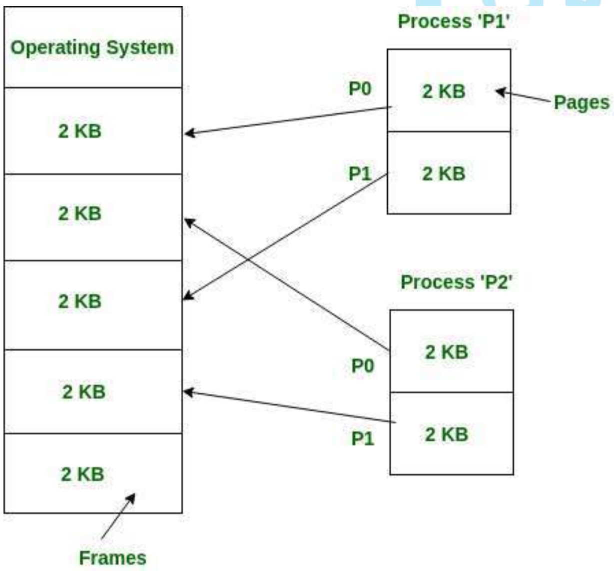
Example: Suppose (consider the above example) three processes p1, p2, and p3 come with sizes 2MB, 4MB, and 7MB respectively. Now they get memory blocks of size 3MB, 6MB, and 7MB allocated respectively. After allocating the process p1 process and the p2 process left 1MB and 2MB. Suppose a new process p4 comes and demands a 3MB block of memory, which is available, but we can not assign it because free memory space is not contiguous. This is called external fragmentation.

- Both the first-fit and best-fit systems for memory allocation are affected by external fragmentation. To overcome the external fragmentation problem Compaction is used. In the compaction technique, all free memory space combines and makes one large block. So, this space can be used by other processes effectively.
- Another possible solution to the external fragmentation is to allow the logical address space of the processes to be noncontiguous, thus permitting a process to be allocated physical memory wherever the latter is available.

Non-Contiguous Allocation in Operating System

Non-contiguous allocation, also known as dynamic or linked allocation, is a memory allocation technique used in operating systems to allocate memory to processes that do not require a contiguous block of memory. In this technique, each process is allocated a series of non-contiguous blocks of memory that can be located anywhere in the physical memory.





➢ Paging

Paging is a technique that eliminates the requirements of contiguous allocation of main memory. In this, the main memory is divided into fixed-size blocks of physical memory called frames. The size of a frame should be kept the same as that of a page to maximize the main memory and avoid external fragmentation.

Advantages of paging:

- Pages reduce external fragmentation.
- Simple to implement.
- Memory efficient.
- Due to the equal size of frames, swapping becomes very easy.
- It is used for faster access of data.

What is Segmentation?

Segmentation is a technique that eliminates the requirements of contiguous allocation of main memory. In this, the main memory is divided into variable-size blocks of physical memory called segments. It is based on the way the programmer follows to structure their programs. With segmented memory allocation, each job is divided into several segments of different sizes, one for each module. Functions, subroutines, stack, array, etc., are examples of such modules

File Management

A file system is a method an operating system uses to store, organize, and manage files and directories on a storage device. Some common types of file systems include:

1. **FAT (File Allocation Table):** An older file system used by older versions of Windows and other operating systems.

   The File Allocation Table (FAT) is a file system structure used by some operating systems, including older versions of Windows. It uses a table to keep track of the allocation status of each cluster (a fixed-size block of storage) on the disk. The FAT file system has evolved over time, with variations such as FAT12, FAT16, and FAT32, supporting different disk sizes and features.
2. **NTFS (New Technology File System):** A modern file system used by Windows. It supports features such as file and folder permissions, compression, and encryption.

NTFS (New Technology File System) is a file system introduced by Microsoft with Windows NT. It is the *default file system used by modern versions of Windows, including Windows XP, Windows 7, Windows 10, and Windows Server editions. NTFS offers features such as improved performance, reliability, security, support for large file sizes and volumes, file compression, encryption, and access control.*

3. **ext (Extended File System):** A file system commonly used on Linux and Unix-based operating systems.
4. **HFS (Hierarchical File System):** A file system used by macOS.
5. **APFS (Apple File System):** A new file system introduced by Apple for their Macs and iOS devices.

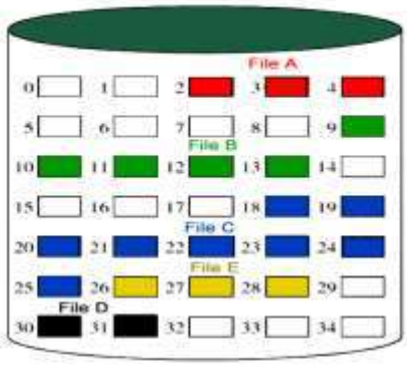
Files Attributes And Their Operations

Attributes	Types	Operations
Name	Doc	Create
Type	Exe	Open
Size	Jpg	Read
Creation Data	Xis	Write
Author	C	Append
Last Modified	Java	Truncate
protection	class	Delete
		Close

FILE ALLOCATION METHODS

Continuous Allocation

A single continuous set of blocks is allocated to a file at the time of file creation. Thus, this is a pre-allocation strategy, using variable size portions. The file allocation table needs just a single entry for each file, showing the starting block and the length of the file. This method is best from the point of view of the individual sequential file. Multiple blocks can be read in at a time to improve I/O performance for sequential processing. It is also easy to retrieve a single block.



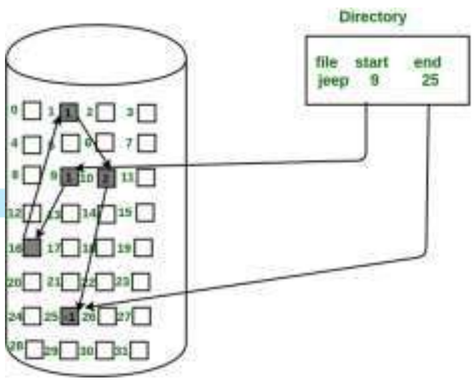
File allocation table

File name	Start block	Length
File A	2	3
File B	9	5
File C	18	8
File D	30	2
File E	26	3

Disadvantage

- External fragmentation will occur, making it difficult to find contiguous blocks of space of sufficient length. A compaction algorithm will be necessary to free up additional space on the disk.
- Also, with pre-allocation, it is necessary to declare the size of the file at the time of creation.

Linked Allocation(Non-contiguous allocation)



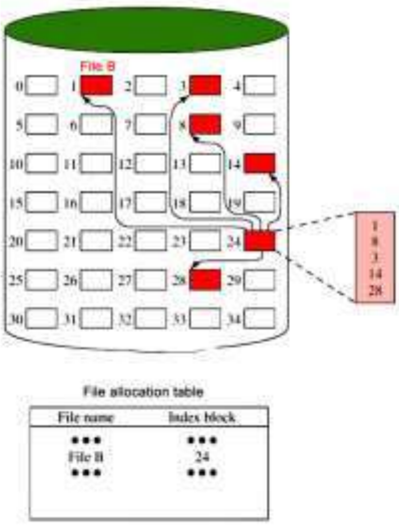
Allocation is on an individual block basis. Each block contains a pointer to the next block in the chain. Again the file table needs just a single entry for each file, showing the starting block and the length of the file. Although pre-allocation is possible, it is more common simply to allocate blocks as needed. Any free block can be added to the chain. The blocks need not be continuous. An increase in file size is always possible if a free disk block is available. There is no external fragmentation because only one block at a time is needed but there can be internal fragmentation but it exists only in the last disk block of the file.

Disadvantage

- Internal fragmentation exists in the last disk block of the file.
- There is an overhead of maintaining the pointer in every disk block.
- If the pointer of any disk block is lost, the file will be truncated.
- It supports only the sequential access of files.

Indexed Allocation

It addresses many of the problems of contiguous and chained allocation. In this case, the file allocation table contains a separate one-level index for each file: The index has one entry for each block allocated to the file. The allocation may be on the basis of fixed-size blocks or variable-sized blocks. Allocation by blocks eliminates external fragmentation, whereas allocation by variable-size blocks improves locality. This allocation technique supports both sequential and direct access to the file and thus is the most popular form of file allocation.



The advantages of using a file system

1. **Organization:** A file system allows files to be organized into directories and subdirectories, making it easier to manage and locate files.
2. **Data protection:** File systems often include features such as file and folder permissions, backup and restore, and error detection and correction, to protect data from loss or corruption.
3. **Improved performance:** A well-designed file system can improve the performance of reading and writing data by organizing it efficiently on disk.

Disadvantages of using a file system

1. **Compatibility issues:** Different file systems may not be compatible with each other, making it difficult to transfer data between different operating systems.
2. **Disk space overhead:** File systems may use some disk space to store metadata and other overhead information, reducing the amount of space available for user data.
3. **Vulnerability:** File systems can be vulnerable to data corruption, malware, and other security threats, which can compromise the stability and security of the system.

Windows OS History:

The history of Windows **dates back to 1981 when Microsoft started work on a program called "Interface Manager."** It was announced in November 1983 (after the Apple Lisa, but before the Macintosh) under the name "Windows," but Windows 1.0 was not released until November 1985.